

PROOF OF CONCEPT | SYNTHETIC DATA

T2D Digital Twin



Predicting 2-hour-ahead hyperglycemia risk by fusing EHR, glucose and wearable data

Digital Twin Challenge 2026

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Team RuntimeRebels

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Glucose problems are usually noticed after they happen

1 Reactive care

Type 2 Diabetes needs continuous management, yet high glucose is typically recognised only once it has already occurred, leaving little room to act early.

2 Data without foresight

Glucose monitors and wearables stream readings around the clock, but raw readings alone do not say what is likely to happen next.

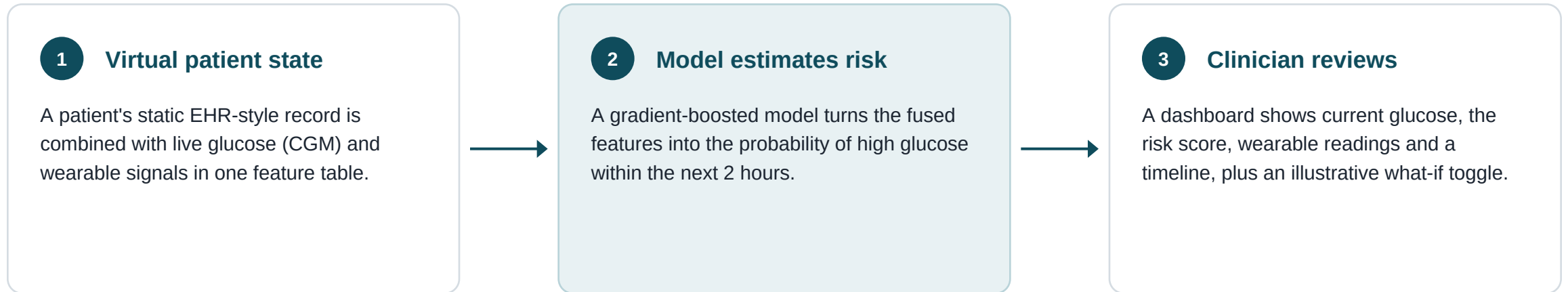
3 The question

Can a patient's history plus live signals flag a hyperglycemic event (glucose above 180 mg/dL) about 2 hours before it happens?

Why prediction could help

An early risk flag could give a clinician or patient time to review the situation and respond before glucose rises. This project is a proof of concept of that idea, built entirely on synthetic data.

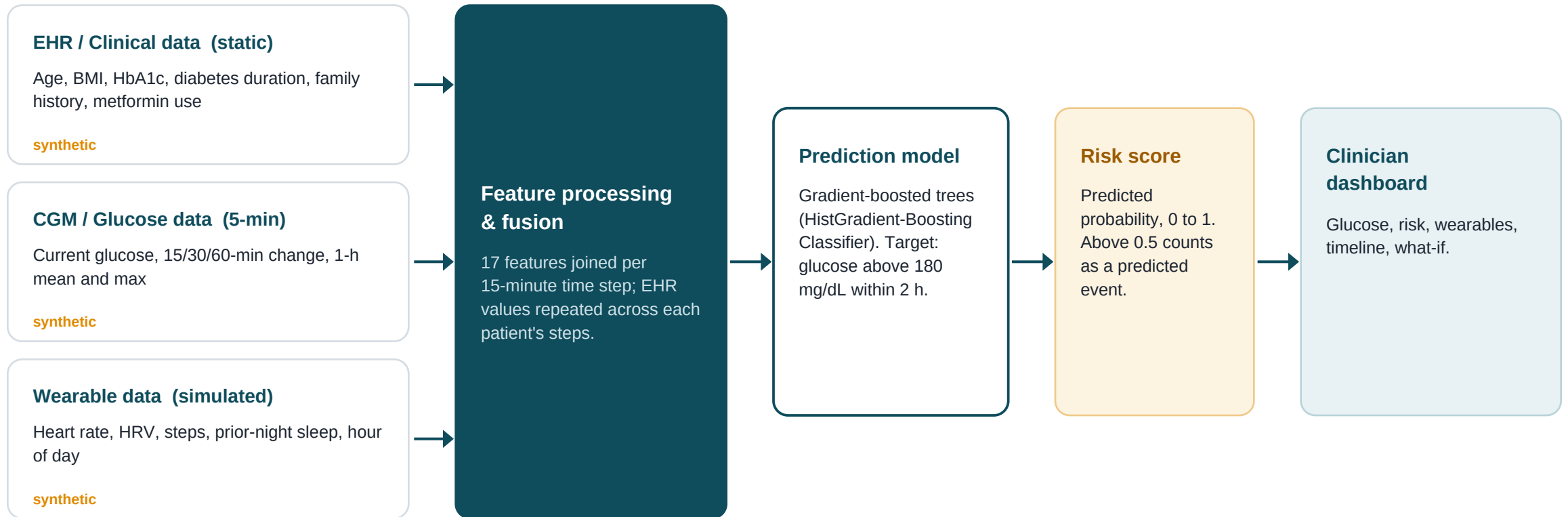
A digital twin that fuses static and live data



In this proof of concept

- Three data streams: static EHR-style fields (6), glucose features (6), wearable features (5)
- Same pipeline run with 1, 2 and 3 streams to measure what each adds
- All data is synthetic; nothing here uses real patient records

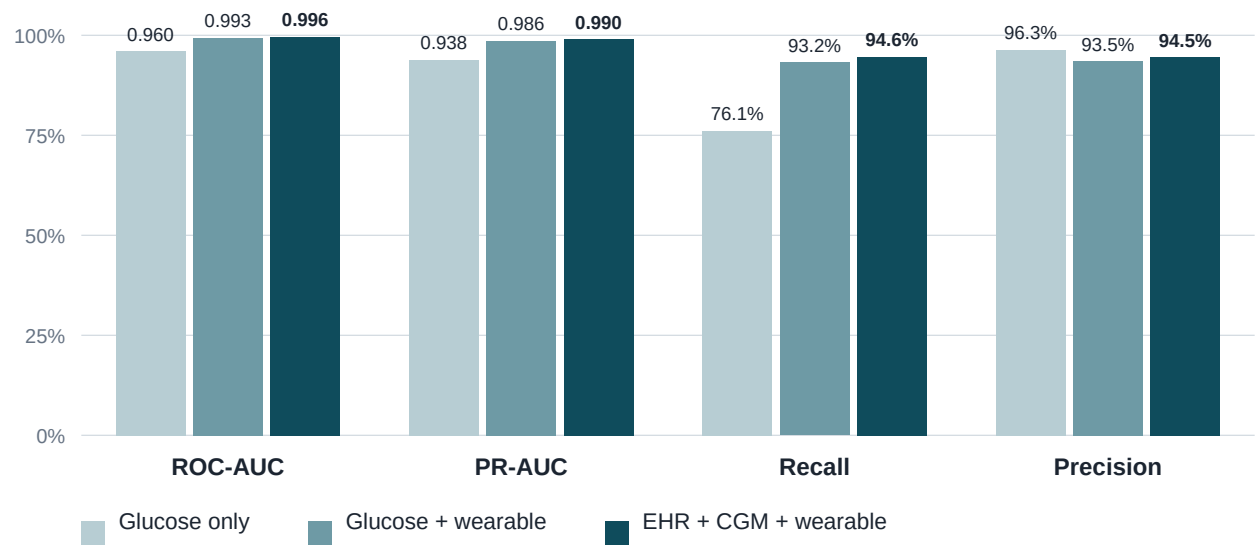
From three data streams to a risk score



Setup: 300 synthetic patients | 14 days each, 5-minute sampling | training rows every 15 minutes | patient-level split, 25% of patients held out for testing

Adding data streams improves detection

HistGradientBoostingClassifier | held-out test: 75 synthetic patients, 99,900 rows, 30.6% events | threshold 0.5



RECALL

+18.5 points

76.1% to 94.6%

ROC-AUC

+0.036

0.960 to 0.996

PR-AUC

+0.052

0.938 to 0.990

PRECISION

1.8 points lower

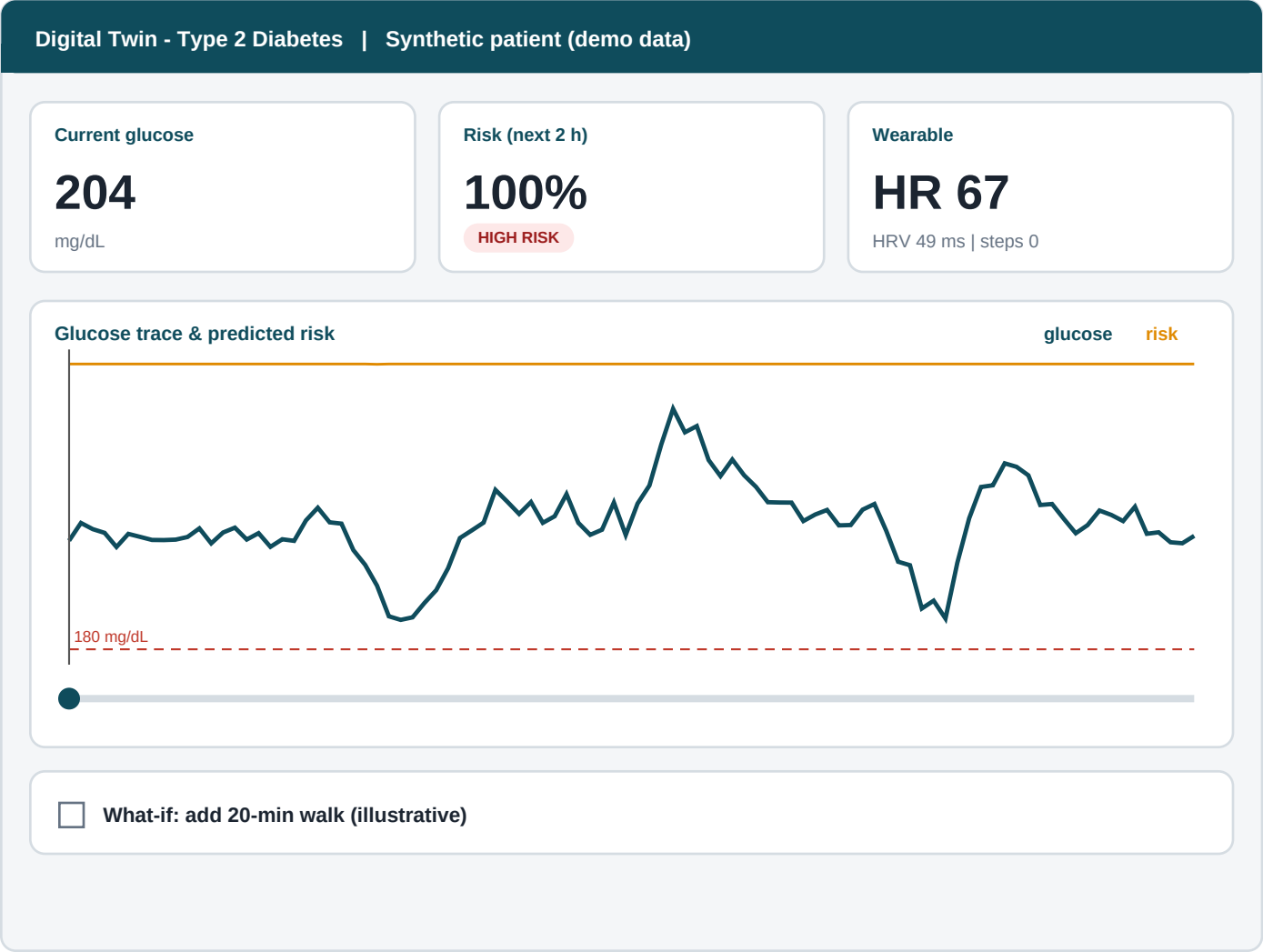
96.3% to 94.5%

Glucose only to fused EHR + CGM + wearable

Model inputs	ROC-AUC	PR-AUC	Recall	Precision
Glucose only	0.960	0.938	76.1%	96.3%
Glucose + wearable	0.993	0.986	93.2%	93.5%
EHR + CGM + wearable (fused)	0.996	0.990	94.6%	94.5%

Scores come from the project's own synthetic data and show the pipeline works end to end; they are not real-world performance.

What a clinician sees



IN THE DASHBOARD

- **Current glucose**
reading at the selected time step
- **Predicted risk**
model probability for the next 2 hours, with a HIGH / LOW flag at 0.5
- **Wearable readings**
heart rate, HRV and recent steps
- **Glucose and risk timeline**
24 hours, scrubbed with a slider
- **What-if toggle**
shows an illustrative fixed risk reduction for a walk (selecting it takes 100% to 65% here)

Demo scope: a static HTML page that replays risk scores precomputed for one synthetic test patient. The what-if toggle is a fixed adjustment, not a model re-run.

What this proof of concept is not

- **Synthetic data only**

Every record and signal is generated by the project's own simulator. No real patient data was used.

- **No clinical validation**

No clinician review, clinical study or external validation has been performed.

- **Results reflect the generated test data**

Model scores were measured on held-out synthetic patients and do not indicate real-world performance.

- **Real-world use needs more validation**

Deployment would require validation on real clinical data, safety review and regulatory consideration.

- **Not medical advice**

This prototype must not be used for diagnosis or treatment decisions.

Realistic paths from prototype to evidence

1

Validate on real clinical datasets

Test on de-identified real-world EHR and CGM data, under the required approvals.

2

Improve personalization

Adapt predictions to each patient's own history instead of one global model.

3

External validation

Evaluate on data from sources not used during development.

4

Clinician evaluation

Review usefulness, interpretability and workflow fit with clinicians.

5

Robust longitudinal modeling

Model longer patient histories over time rather than short fixed windows.

6

Privacy, security and deployment

Plan data protection and secure deployment before any real-patient use.